Supplementary Material for TerraState: A Testable Predictive-State World Model for Weather-Driven Land-Surface Forecasting

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This technical appendix provides details omitted from Sections 3 and 4 of the main paper. Appendix A specifies implementation details of the predictive state, transition, readout, and training targets; Appendix B records the training, preprocessing, and scoring protocol; and Appendices C–D give the complete Q2 and Q3 intervention procedures. Numerical Q1–Q3 results remain in the main paper to avoid duplicating its tables.

A Additional Implementation Details

A.1 Context Isolation and Predictive-State Construction

At forecast issue time t, the optical context is $\mathcal{H}_t = \{(x_i, m_i, \tau_i)\}_{i=1}^{C}$, where x_i is a Sentinel-2 composite, m_i is its validity mask, and τ_i is the acquisition time. Historical meteorology is denoted by $u_{\leq t}^{\text{past}}$, the ordered future meteorological sequence by $u_{t+1:t+H}$, static geography by g, and the future optical targets by $y_{t+1:t+H}$. The context $\widetilde{\mathcal{C}}_t = (\mathcal{H}_t, u_{\leq t}^{\text{past}}, g)$ contains neither future optical observations nor future meteorology.

Each minicube contains 30 five-day composites at 128×128 pixels. The first $C = 10$ composites form the context, and the following $H = 20$ composites form the 100-day prediction window. Optical input has five channels, including normalized difference vegetation index (NDVI) and four reflectance bands. Meteorology has 24 channels, obtained from eight variables with mean, minimum, and maximum aggregation. Static input has five channels; the first three elevation products provide the patch-wise geography input.

The history operator uses the PVT v2/Contextformer configuration adopted by TerraState. A 4×4 patch size yields $32 \times 32 = 1{,}024$ spatial tokens for each minicube. The projector applies layer normalization, a two-layer multilayer perceptron, and output normalization to the final-context token at every patch. The resulting per-minicube state has shape $[1024, 256]$, or $[1024B, 256]$ after flattening the batch and patch dimensions.

The context-only constraint is structural. Future optical tensors and future meteorology are zeroed before the history pass, and future image positions are masked by the history operator. Consequently, $b_{1:H}$ and z_t depend only on historical observations, past meteorology, and static geography. This separation makes the later weather substitution specific to the state transition.

A.2 Direct Shared Transition and State Readout

For each queried horizon, the shared transition is conditioned on the corresponding ordered future-weather prefix. A shared gated recurrent unit produces codes for all 20 prefixes in one recurrent pass. A geography encoder average-pools the three elevation channels to the 4×4 patch grid, and a sinusoidal embedding represents elapsed horizon. The fusion network and residual transition share parameters across patches and horizons; geography remains patch-specific, whereas weather and horizon codes are broadcast across the patch grid.

Every horizon is queried directly from the same history-derived state z_t. The model applies the shared transition once for each h, using the prefix $u_{t+1:t+h}$, rather than feeding z_{t+h-1} recursively into the next horizon. The state readout maps each 256-dimensional transitioned token to one local 4×4 output patch and unpatchifies the patches into $r_h \in \mathbb{R}^{1 \times 128 \times 128}$. The coefficient multiplying this contribution is a fixed non-learnable buffer with value one, which supplies the precise Q2 cut point.

A.3 Training-Target Construction

The ground-truth objective first averages squared error over clear horizons for each pixel and then averages over eligible vegetation pixels. A frozen full-weather KD teacher supplies a stopped forecast target, weighted by 0.5 in the complete objective.

The future-state objective anchors only the terminal transitioned state. A training-start frozen copy of the history operator and projector processes the observed all-frame optical sequence with recorded masks while future weather is zeroed. The patch mask requires a fully clear terminal 4×4 patch containing at least one vegetation pixel. Layer-normalized student and target tokens are compared by cosine distance. Future optical observations create this stopped training target only; they are absent from the deployed inference graph. The teacher and target encoder are discarded after training, and Q2–Q3 introduce no additional objective.

Setting	Value
Epochs / optimizer updates	40 / 14,880
Optimizer	AdamW
β_1, β_2 / weight decay	$0.9, 0.999$ / 0
Branch / history learning rate	3×10^{-5} / 9.9×10^{-7}
Learning-rate schedule	300-update warm-up, cosine decay
Global batch	64 (8 devices × 8 samples)
Gradient accumulation	none
Gradient clipping	global norm 1.0
Precision	FP32
Hardware	8 NVIDIA H200 GPUs

Table 1: TerraState training configuration.

B Training and Evaluation Protocol

B.1 Training and Implementation Details

Table 1 gives the TerraState training configuration. Training lasts 40 epochs and exactly 14,880 optimizer updates. The implementation uses AdamW with $\beta = (0.9, 0.999)$, zero weight decay, and gradient clipping at norm 1.0. The non-history branch uses a base learning rate of 3×10^{-5}. The history-operator parameter group uses 9.9×10^{-7}, equal to 0.033 times the branch rate.

The update schedule controls both trainable parameters and λ_s. During the first 20% of updates, the history operator remains frozen and λ_s increases linearly from 0 to 0.02. From 20% through 80%, the history operator remains frozen and $\lambda_s = 0.02$. During the final 20%, only the final Transformer block of the history operator is unfrozen, and $\lambda_s = 0.01$. The projector, weather encoder, geography encoder, fusion network, transition, and readout are optimized throughout.

The global batch is 64, realized as eight samples on each of eight GPUs without gradient accumulation. The shared learning-rate multiplier increases linearly during the first 300 optimizer updates and then follows cosine decay, preserving the ratio between the two parameter-group rates.

Model selection uses validation forecasting performance only. Neither Q2, Q3, nor OOD-t contributes to selection. The selected model is evaluated without parameter updates on all three questions, thereby keeping the intervention results outside checkpoint selection and parameter updates.

B.2 GreenEarthNet Splits and Preprocessing

GreenEarthNet supplies separate training, validation, and temporal out-of-distribution partitions. Training optimizes the model, validation supports checkpoint selection and the validation Q2 analysis, and OOD-t is held out from selection. The OOD-t evaluation contains 1,904 minicubes. The validation Q2 scorer contains 952 target minicubes, of which 589 yield paired metric units after the scorer's eligibility rules.

For each minicube, the data adapter samples the optical record every five days to obtain 30 composites. NDVI is computed from near-infrared and red reflectance. Missing optical values are set to zero after a separate cloud and quality mask is constructed. The cloud mask combines the provided learned mask with valid scene-class labels. Meteorological variables are normalized using the GreenEarthNet statistics and aggregated over each five-day interval by mean, minimum, and maximum, producing 24 channels. Elevation channels are scaled by 500, and categorical static layers retain their original scale.

The forecast target is NDVI over valid vegetation pixels. Clear target observations satisfy the official cloud and scene-class rule, and vegetation pixels fall in the declared land-cover range. Prediction tensors have shape $[B, 20, 1, 128, 128]$. For file-based scoring, each output stores the 20 NDVI predictions on the target timestamps with the original latitude and longitude coordinates. The official scorer computes pixel-level time-series quantities before balancing them across minicubes and vegetation land-cover classes. It reports R^2, RMSE, NSE, absolute bias, and RMSE_{25}, as defined under *Experimental Setup* in Section 4 of the main paper.

C Q2 Intervention and Statistical Protocol

Q2 evaluates three forward computations on identical samples. The full model uses the additive state path in the main paper. State removal sets the state contribution to zero immediately before addition, giving $\widehat{y}_{t+h}^{\mathrm{remove}} = b_h$. The identity-transition control replaces the learned transition output by the unevolved z_t while keeping the state readout active. Model weights, history, target window, masks, and scorer remain fixed for every arm.

The analysis reports two distinct estimands. Official ΔR^2 is the difference between full and intervened dataset-level scores. Paired ΔR^2 is computed per eligible minicube and then averaged; its 95% percentile interval comes from 10,000 paired bootstrap resamples. Because the two quantities use different aggregation orders, the paired interval characterizes the mean paired-minicube effect rather than the official dataset-level difference.

The state-removal comparison provides the primary evidence: the explicit state contribution is removed without changing the matched context forecast. The identity-transition arm is a supporting diagnostic because it can present the readout with states outside its trained input distribution. Numerical effects for both splits are reported in Table 2 of the main paper.

Implementation invariants check that all arms use identical sample order, target tensors, masks, history-derived context forecast, and frozen model weights. They also verify that state removal recovers b_h, the fixed state-contribution coefficient is restored after each call, and finite paired units are aligned before bootstrapping.

D Q3 Heat–Drought Subset and Weather-Control Protocol

Q3 uses a frozen heat–drought subset with 84 matched pairs. The 84 evaluation samples comprise all eligible OOD-t minicubes in the predeclared heat–drought stratum. Hot and dry anomaly thresholds were estimated from the training climatology at the 80th percentile and frozen before model evaluation; eligible samples additionally required at least 80% valid future-weather coverage. Neither future NDVI, model predictions, forecast errors, nor checkpoint outputs entered subset construction. For each sample, the donor was selected from the normal-weather pool under an exact seasonal constraint and matching on geography, historical context, vegetation coverage, and data quality. This frozen procedure produced 84 pairs involving 45 unique donors and 31 geographic clusters.

For the weather intervention, each evaluated minicube retains its historical context, $b_{1:H}$, z_t, geography, horizon queries, readout, target window, and mask. Only the future-weather tensor supplied to T_ψ changes among three arms:

1. actual future weather from the evaluated minicube;
2. donor future weather from the frozen matching procedure;
3. normalized-mean weather, represented by zero in the frozen global standardized feature space.

The primary loss is masked mean squared error over the complete 20-step forecast window. Control-minus-actual loss is computed per pair, so a positive value favors actual weather. The uncertainty analysis resamples the 31 geographic clusters with replacement for 10,000 replicates, preserving dependence caused by donor reuse. Whether actual weather yields lower loss is summarized by a descriptive pair count, while statistical inference relies on the geographic-cluster bootstrap. The subset R^2 and RMSE in Table 3 of the main paper refer specifically to these 84 matched samples.

This intervention measures conditional response fidelity under a frozen matched protocol. This estimand characterizes response fidelity under observed targets; causal and counterfactual identification lie outside its scope because alternate real-world outcomes under the control weather sequences are unobserved.